

Task - 2

Problem 2.A

Case ①

let exactly k contribute & $n-k$ not contribute.

Then, good is provided.

If contributor switches to not contributing → outcome changes from
no advantage in switching

if non-contributor switches to contributing
no advantage in switching → outcome changes from

$2 \rightarrow 3$
worse

∴ Nash equilibrium. → exactly k contribute.

$1 \rightarrow 2$
worse.

Case ②

let $< k$ contribute } no public good provided.

If a contributor switches to not contributing →
Not Nash

outcome
 $4 \rightarrow 3$
better

Case ③

let $> k$ contribute } public good provided.

If a contributor switches to not contrib...
Not Nash

outcome
 $2 \rightarrow 1$
better

Nash equilibrium is when exactly k contribute & $n-k$ don't.

Problem 2. B.

Case

① a portion of stag is always preferred over hare.

① Then let's say $\geq m$ are catching stag.

Then if one catching hare goes to catch stag,
payoff increases for him $\geq$
Not Nash.

② if $< m$, then, hunter catching stag will prefer to catch hare rather than getting nothing
Not Nash

No Nash equilibria.

Case ② if $> k$ hunters ($m \leq k \leq n$) hunters hunt stag, they would rather prefer hare.

① Let $> k$ hunters go for stag, they would prefer hare. (Not Nash).

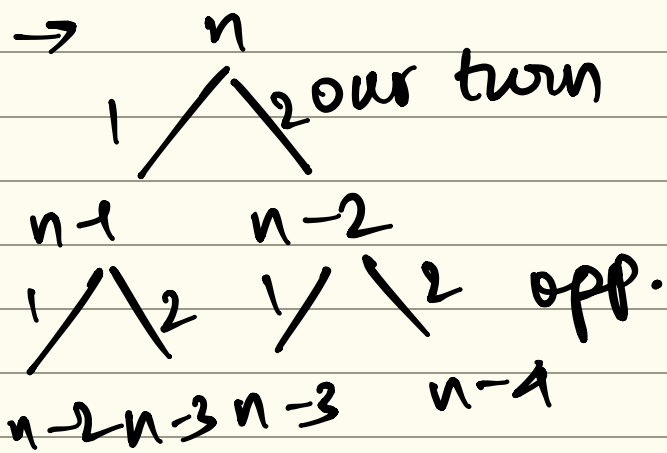
② Let exactly k hunters go for stag
→ stag hunters don't switch since they (Nash) prefer $1/k$ of stag over hare
→ hare hunters don't switch because they prefer hare over $\frac{1}{k+1}$ of stag

③ If a hunters, ($m \leq a < k$) go for stag, hare hunters would prefer to switch (Not Nash)

④ If $< m$ hunters go for stag, stag hunters would prefer to switch (Not Nash)

[Nash eqm. = exactly k hunt stag.]

Task -3



1 → take from 1st pile
2 → take from 2nd pile
3 → both.

1, 2 are equivalent

∴ The choices are

1 → take 1
2 → take 2.

→ opp can always make
 $n \rightarrow n-3$ no matter
what we play.

∴ As $n=20$ at start.

opp takes 2, $n=18$.

Then opp can always get $n-3$ at end
of turn no

∴ $18 \rightarrow 15 \rightarrow 12 \rightarrow 9 \rightarrow 6 \rightarrow 3 \rightarrow 0$ matter what
we play.

2
opp wins.

∴ Opponent has guaranteed win,
he takes 2 stones,

Then if we take 1, he takes 2 } repeat
& if we take 2, he takes 1 } until
victory